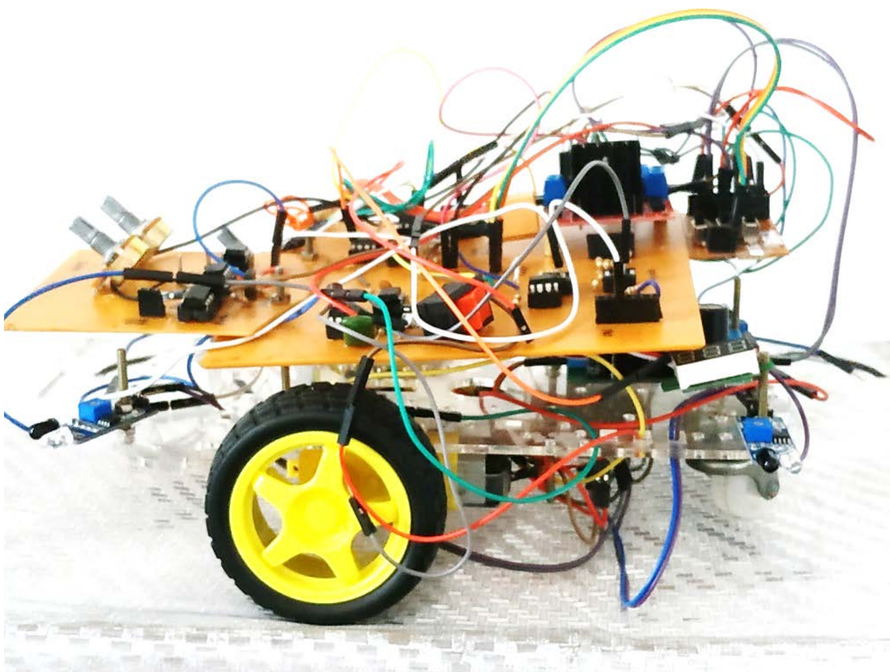
Department of Electronic and Telecommunication Engineering
University of Moratuwa



Autonomous Analog Wall Following Robot

Team Members

N.P.Y Yasawardhana 160721R
K.I.Akalanka 160015G
T.T.N Bahavan 160050H
R.P.Ekanayake 160145E

This is submitted as a partial fulfillment for the module EN2090: Laboratory Practice II
Department of Electronic and Telecommunication Engineering
University of Moratuwa

13th July 2018

Executive Summary

Problem

The task was to create an analog wall follower, which is a robot that capable of moving along a corridor without colliding with the wall on either side of it and the navigation was to be done by detecting the walls and without using microcontrollers.

The robot should be capable of guiding itself along the walls a 400mm wide corridor and be able to navigate curved bends. There should be zero user interference. Additionally, the robot should be able to detect walls on both sides and to have a Proportional Integral Differential (PID) control system for handling itself.

Design Description

The robot comprises of three wheels, of which one is an omnidirectional wheel which was included for stability. The steering mechanism was achieved by independent control of the speed of the two actively rotating wheels.

The wall follower that was designed to use an Infrared (IR) Light emitter and detector combination to detect the wall. The design was to have 4 sensors, two on either side of the robot to detect the wall on both sides. The readings from one side were considered as negative and the other side was taken to be positive. The net signal gave the position of the robot relative to the wall. (The weightage of a sensor was depending on the position of that sensor relative to the two motors.)

The net signal was passed on to a PID control system and the resulting output was compared with a triangular wave. The comparison created a Pulse Width Modulated (PWM) voltage signal that was added to a constant voltage to create the control signal for one wheel. To create the control signal for the other wheel, the PWM signal was deducted from the same constant voltage. The control signals were then passed on to a motor controller that adjusted the speed of the wheels accordingly.

Evaluation

The design was evaluated using computer models as well as a prototype. The complete circuit was designed as four separate modules named 1) Triangle wave generator, 2) Input Handler, 3) PWM generator and 4) PID. Each sub-circuit was tested separately by providing the theoretical input signals to the components and observing the output. The triangle generator required no input.

The sub-circuits were combined into a single circuit and the combined operation was evaluated. A working model of the circuit was built on prototype boards before the printed circuit board (PCB) was made and this was evaluated. The same tests were carried out on the PCB. The robot was found to operate as intended.

The wall follower operated as intended albeit a bit too faster than it can process the sensor inputs. Therefore, it was necessary to reduce the speed of the motors by reducing the current passing through each motor. The additional circuitry made it possible to equate the rpm of two different rpm motors. Even though the robot did not collide whilst navigating the corridor it did not always maintain a position equidistant from the two walls. Improvements can be made in these areas.

Table of Contents

Introduction

A Wall Follower is an autonomous robot that guides itself alongside a wall without colliding with the wall. The analog wall follower does not use any microcontrollers for handling itself. This was designed using analog circuit components such as op-amps, resistors, capacitors, etc.

This report gives the reader a comprehensive description of the project, the desired and achieved learning objectives. It also contains the datasheets for the component.

Problem Definition

Problem Scope

To design a robot capable of navigating a corridor from the start point to the end, without user intervention, without colliding with the walls on either side of the robot. Navigation should be done by detecting the walls and the robot should not use any micro-controllers.

Technical review

Field of invention

This invention relates to autonomous robots which use wall-following method of navigation and a sensor subsystem defined with respect to the walls. More specifically to strictly hardware defined autonomous robots that use the wall following method of navigation.

Wall following is primarily used for indoor robotics and household appliances. Currently, indoor drones are being developed so that it can navigate via the walls without using technologies such as GPS. There are many potential usages for these autonomous wall-following technologies such as the searching of a burning building or one suspected of explosive rigging or anti-terrorism activities.

Description of Related Prior Art

Evolution of the wall following robots has started from the robots with single distance sensor to keep a mobile robot at a determined distance to a reference surface while moving forward with constant velocity. For comparison purposes, the relative position to the reference surface is first determined from two measurements which is then extended to use just one sensor.

magnetic track drive robot or the Modular Inspection Crawler is another invention which its magnetic wheels allow a broad range of applications, including driving up on a vertical wall or overhanging which can be used with one, two or more ultrasonic sensors as a wall follower. Wall Following with Collision Avoidance and Mapping using a laser range finder, automatic navigating wall following robot used in various real-world tasks such as underwater exploration, unmanned flight, and in automotive industries are some of the autonomous wall followers built so far.

Our project was to build an analog wall following robot which was far more cost efficient compared to the above-mentioned inventions. Since microcontrollers were not needed and IR sensors were used instead of sonar sensors, we could build our analog wall follower with a minimum cost.

Design requirements

The design requirements are extracted from the "Project Description" document provided by the project supervisor. Further requirements were verbally communicated at project meetings.

Requirement	Importance	Units	Marginal Value	Ideal Value
Proximity Sensors	Prevent collision and detect the wall on both sides	number of units	2	4
Robot Width	Corridor to navigate is only 400mm wide	millimeters	400	200
Sensor Placement	Should be placed less 200mm above ground level	millimeters	200	Less than 200mm
Control System	Smooth navigation			
User intervention	Autonomy	instances	0	0

<u>Proximity Sensors</u>

Proximity sensors are sensors that can detect the distance between itself and another object. These sensors can use different mechanisms to measure the distance.

The use of proximity sensors is derived from the instruction in the project description document that requires the robot to navigate by detecting the walls.

The proximity sensors are needed to maintain a specific distance between the wall and the robot. This helps to prevent the robot from colliding with the walls on either side.

<u>Robot Width</u>

The robot should have a width that allows it to enter the corridor as well as navigate the curved bends.

A restriction on the width of the robot is imposed as the "Project Description" document describes that the width of the corridor to be navigated is 400mm. This sets an upper limit on the width of the robot at its widest point.

The width should be less than 400mm so that the robot may successfully navigate the corridor. The ideal width will allow the robot to simultaneously detect the wall on both sides of the robot.

<u>Sensor placement</u>

The placement of the sensors is how the sensors are arranged around the robot.

Sensor placement is determined by the problem requirement to detect the wall on both sides of the robot and by the fact that the walls are only 200mm tall.

The placement determines the behavior of the robot. Also correct placement is needed to detect the walls without cross interference between the sensors. The sensors should be placed at a height that can detect the wall.

Control System

Control system is the system of devices that manages the behavior of the devices using control loops. (electrical4u, 2018)

"Project Description" document lists a PID control system as an evaluation criterion. This implies the requirement for a PID control system.

The control system allows for faster operation of the robot. Having a PID control system allows the robot to adjust itself quicker.

User intervention

During the operation of the robot, it may be unable to proceed with the task and will need help from the operator to proceed. This is defined as user intervention.

The problem requires an autonomous solution. Therefore, there is a requirement for zero user intervention.

Design Description

Overview

The designed robot obtains data about its orientation with respect to the adjacent walls from its four sensors and feeds it to the sensor fusion unit, which converts it to a single signal. This signal is processed by the PID and fed into the PWM unit along with the triangle wave generator output. The PWM unit then adjusts the speed of the two active motors via the motor-controller so that the robot avoids colliding with the wall and follows it.

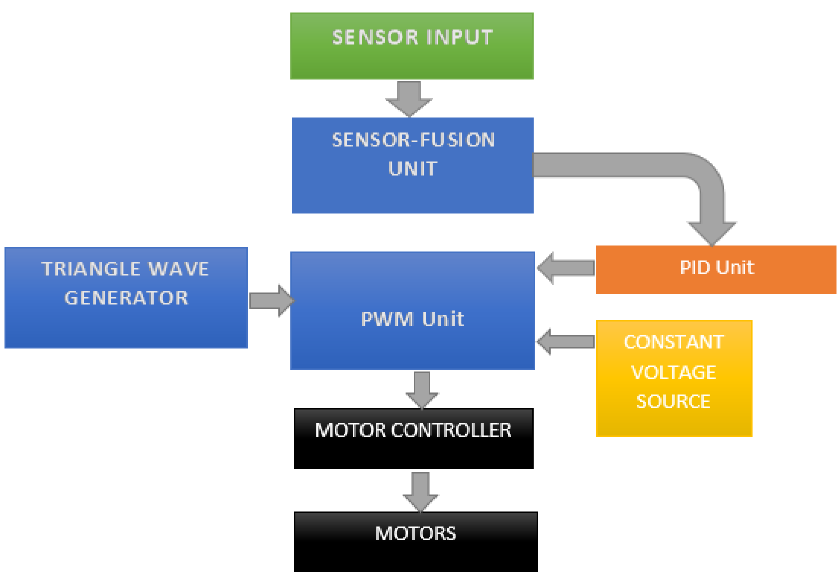


Figure 1: Overview of Operation

General Specifications of the Robot

Weight	Height	Width	No of motors	No. of wheels	Power Source and Voltages	No. Sensors
430 grams (with battery)	154mm	130mm	2	3	LIPO (11.1V, 1600MAH); Batteries.	4

Printed Circuit Board (PCB)	Nature of PCB	Dimens ions	No. Of Jump ers	Power Specifications	No. of ports	Contained Units
Main Board	Single Sided	127mm x 127mm	7	11.1v, 4.85v, 0v	16	Sensor Fusion unit, PWM Unit, Triangle wave generator
PID Unit	Single Sided	85mm x85mm	2	10v , 0v, -10v	5	PID unit
SWITCH	VERO Board	75mm x60mm	0	10v, 4.85v, 0v	6	Constant Voltage source
Motor Controller	Double Sided	45mm x45mm	0	11.1v, 0v	13	
Voltage regulator	Single Sided	60mm x45mm	0	11.1v (input), 4.85v (adjustable output)	12	
Sensors	Double Sided	30mm x10mm	0	g	3	1x IR emitter and detector couple

Sensors

Introduction

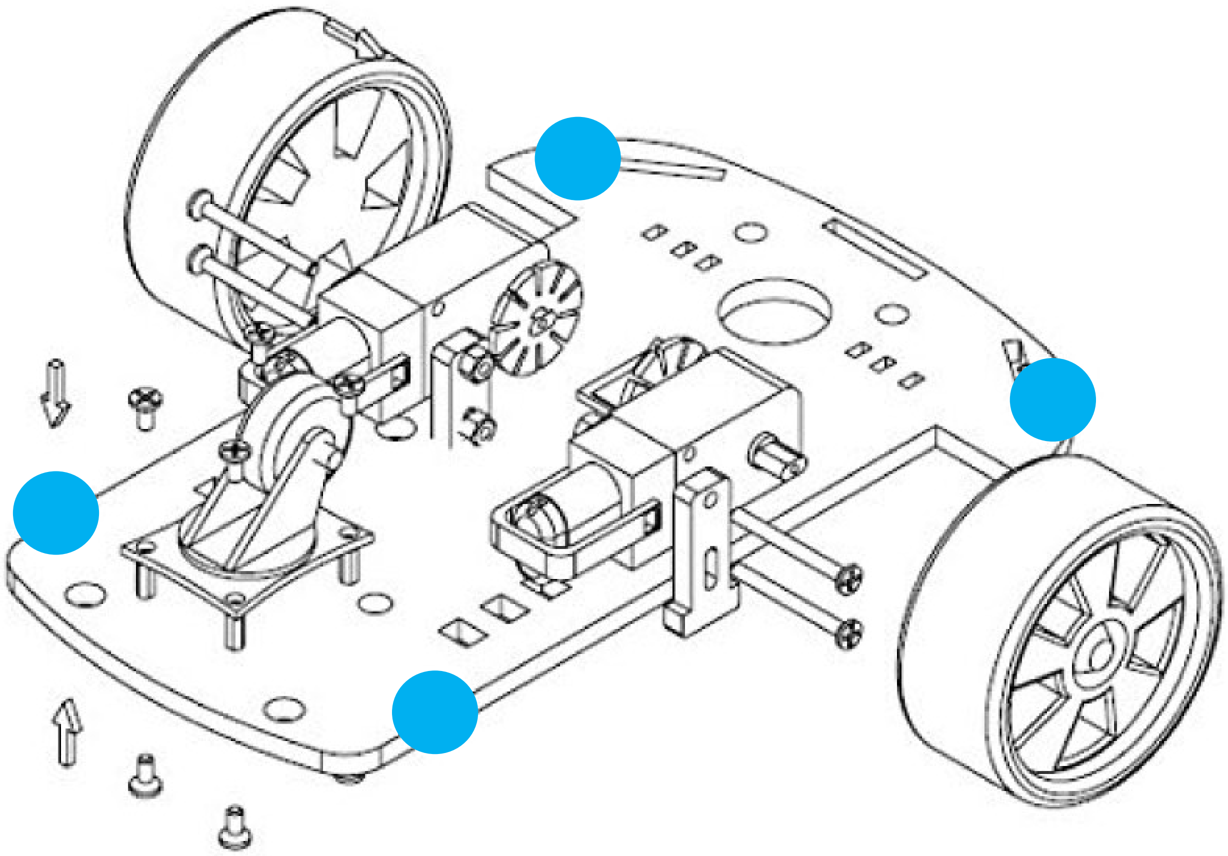
Obstacle avoiding IR detecting sensors (Model: FC-51 (refer appendices)) which can detect objects at a distance in a range between 2~30cm.

Working Principle

The package is an infrared system consisting of an IR emitter (IR LED) and an IR receiver (IR photodiode). IR emitter of the IR module emits IR in the frequency range of infrared, invisible to the naked eye which gets reflected by the obstacles. When the reflected IR light is detected by the IR receiver, an analog voltage is generated proportional to the distance to the obstacle. Since an analog output signal is required, this analog output is taken by short-circuiting the sensor module.

Design

Four sensors were placed on top right, top left, bottom right and bottom left corners of the robot.



Sensor Positions

Rationale

When the robot is negotiating a bend, the corners of the robot are the parts most susceptible to collision. Thus, four sensors were chosen as it was the most cost-effective, reliable and power effective solution.

IR sensors were preferred over Ultra sonic sensors due to their faster response time, narrower beam width, and lower cost.

Sensor Fusion Unit

Introduction

Sensor fusion unit combines all the sensor inputs into one signal.

Working Principle

By combining all the sensor inputs into one signal, it makes it easier to process the signal and this simplifies the circuit. The combined signal gives the instantaneous orientation of the robot with respect to the two adjacent opposing walls.

Design

This consists of two Op-amp adders and Op-amp differential amplifier. The top right and bottom left signals were added together and subtracted from the sum of the top left and bottom right signals.

Rationale

When a sensor detects a wall, the objective of the robot is to steer the robot away from the wall. Consider the case of the top right and bottom left sensors. When the top right sensor is activated, the robot should turn anticlockwise. The same applies when the bottom left sensor is activated. Thus, these signals are added together, as they require similar operations. However, when the other two sensors are activated, the robot should turn clockwise; thus, this signal should have the opposite effect. The difference of the two signals determine what way the robot should turn. Suppose if the difference is null, then the robot is safe from collisions and should proceed along the straight path.

	Top Right	**Top Left**	**Bottom Right**	**Bottom Left**
Adder Resistor Value	10k	10k	8.2k	8.2k
Adder Unit	Right	Left	Left	Right
Distance from the Motor axis				

PID Control Circuit

Introduction

A proportional–integral–derivative controller (PID controller) is a control loop feedback mechanism requiring continuously modulated control. A PID controller continuously calculates an error value setpoint (SP) and a measured process variable (PV) and applies a correction based on proportional, integral, and derivative terms (denoted P, I, and D respectively).

PID is a control system which continuously calculates an error value setpoint and a measured process variable and makes a correction based on the integral, proportional and derivative terms.

Working Principle

The overall control function can be expressed mathematically as

$$u(t) = K_p e(t) + K_i \int_0^t e(t')dt' + K_d \frac{de(t)}{dt}$$

where K_p, K_i and K_d are all non-negative coefficients

The motor current is set in proportion to the existing error by the proportional term. Integral term increases action in relation not only to the error but also the time. The derivative term does not consider the error but tries to bring the rate of change of error to zero.

Design

The PID unit takes in the sensor fusion signal, then feeds it into an Op-amp Differentiator and an Op-amp inverting amplifier; Then the two processed signals are added together via a summing Op-amp. Given that it is possible to adjust the weights of the two signals in a summing Op-amp, two variable resistors were placed so that required K_p, K_d can be calculated while fine tuning the robot. K_i was omitted as it is not necessary for the control system to change the output according to time.

Rationale

The PID smoothens the preprocessed robot position voltage signal so that the robot can handle sharp turns. By adjusting the variable resistors, it is possible to make the robot movement smoother. However, this Robot does not have an Integrator unit in it, as the cost (power) of having it outweighs the benefits it offers for the applications in the design requirements.

Triangular waveform generator

Introduction

Triangle Wave generator was required to convert the sensor fusion signal into a PWM signal, which can be used to control the motors.

Design and Working Principle

Triangular waveform generator is designed using two 741 op amps by cascading an integrator and a square wave generator. First op amp functions as a comparator and the next op amp as an integrator. The astable output from the first Op-amp is fed to the integrator circuit which generates the triangular waveform.

Frequency:	812Hz
V_{pp}:	11.1V

PWM Unit

Introduction

Pulse Width Modulation (PWM) speed control technique which gets analog results with digital means and works by driving the motor with a series of “ON-OFF” pulses and varying the duty cycle which is the fraction of time that the output voltage is active compared to when it is inactive, while keeping the frequency constant. Duration of ‘ON time’ is called the pulse width which can be modulated to obtain varying analog values.

Here the objective of the unit is to convert the PID-processed sensor-fusion signal into two PWM signals for the two respective motors.

Working Principle

The voltage control PWM is initially generated from the triangle signal generator which gives the fundamental PWM pulse frequency and the required ramp voltage.

The voltage control PWM’s pulse frequency and ramp voltage is provided by the triangle signal generator.

By continuously comparing this voltage signal to the voltage signal from the PID system, the required PWM wave is generated.

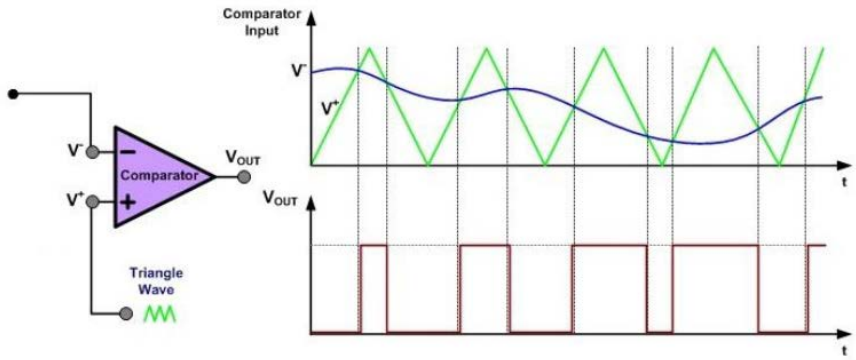


Figure 2: How a PWM Wave is created

Design

After continuously comparing the PID signal with the triangle wave and converting it into a PWM signal, the resulting signal is added with a constant voltage and fed into one motor via the motor-controller; and subtracted from the constant voltage and fed into the other motor via the motor-controller.

Rationale.

Even though the PWM signals fed into the motor-controller are different and inverted, what the motor 'sees' is the average value of the PWM wave. The sum of the averages of the two PWM waves is equal to the maximum voltage that can be supplied (which is a constant). Thus, when inverting one PWM wave, that average value is equal to the difference between the maximum voltage value and the non-inverted PWM wave. Therefore, there is a net difference in between the voltages supplied to the motors which causes the robot to turn. (else the average values would be the same, which will make the robot to go straight)

Motor Controller

Introduction

A Motor Controller is a device that acts as intermediary between the robot's mainboard, batteries and motors. (RobotShop inc., 2018) A motor controller is necessary because a main board can usually only provide roughly 0.1 Amps of current whereas most actuators (DC motors, DC gear motors, servo motors etc.) require several Amps.

A Motor controller acts an intermediary between the robot's batteries, robot's motors and robot's mainboard. A motor controller is necessary as a main board can only provide a small current (0.1A) while most DC gear motors require several Amps.

The current a motor consumes is related to the torque it can provide (a small motor will not consume much current but can't move a large mass whereas a large motor can move a large mass but will consume a large current).

Operating Principle.

The enable pins for each corresponding motor is connected to the PWM unit, thus turning on and off the motor as per the PWM signal to make it move in the required direction.

Design

L298 ready-made motor controller was used.

Rationale

L298 Motor controller can handle more current and voltage than the L293 motor controller (L298 has current capacity of 2A per channel at 45V compared to 0.6 A at 36 V of a L293D). It is also very cheap and modular, and it includes thermal shutdown (it will stop when overloaded thereby preventing melting), indicator LEDs, and a low-drop out regulator.

Motors & Chassis

The Chassis consists of two conventional DC motors along with an omni directional wheel for stability.

Power Supply Unit

Adjustable voltage regulator

Adjustable voltage regulator is a regulator with 3 terminals which has the ability to supply different DC voltage outputs other than the fixed voltage power supply of +5 or +12 volts.

For our robot, 10V voltage supplied by the lithium polymer (LiPo) battery was regulated to 5V from the adjustable voltage regulator and the voltage regulation was done by the IC LM317.

Power to PWM generator:

For the op-amps in the PWM generator, three voltages were used:

11.1V	$+V_{cc}$ supplied by the Lithium Polymer (LiPo)
0V	$-V_{cc}$
5V	Virtual ground taken from the motor controllers regulated 5v output.

Power to Motors:

Motors were fed with 11.1V which was supplied by the Lithium Polymer (LiPo) battery.

Power to the Sensors:

5V regulated from the adjustable voltage regulator was given to all the sensors.

Power to the Motor controller:

11.1V and 0V were supplied to the motor controller from the LiPo battery.

Power to PID:

3 types of voltages +10V,0V and-10V were used for the PID.0V was taken as the ground voltage by short-circuiting the positive terminal of one cell with a negative terminal of another cell. Operating range of the op-amps: –10V to +10V.

Use

The robot is placed on the track. It contains an ON switch for the motor-controller and sensor power supply and another ON switch to activate the main board consisting of the sensor fusion unit, PWM etc. The switches are activated, and the robot will begin its operation.

Evaluation

Overview

The robot was tested via the experimental testing on a cardboard maze. When designing the units, each of them were evaluated by computer simulation on Altium; followed by modelling them on breadboards and testing the breadboard prototypes individually by an Oscilloscope.

Requirement	Importance	Test Method
Proximity Sensors	Prevent collision and detect the wall on both sides	Experimental testing on a cardboard maze with a variety of bends and turns.
Robot Width	Corridor to navigate is only 400mm wide	Experimental testing on a cardboard maze with a minimum corridor width of 400mm
Sensor Placement	Should be placed less 200mm above ground level	Experimental testing on a cardboard maze with a minimum height of 200m
Control System	Smooth navigation	Experimental testing with and without the PID system and comparing the time taken to complete a task for different PID values
User intervention	Autonomy	Experimental Testing on a cardboard maze and obtaining the success rate of its task completion without wall collision.

Prototype

Computer Simulation. The Triangle wave generator was simulated by Altium to see the feasibility of its circuit design.

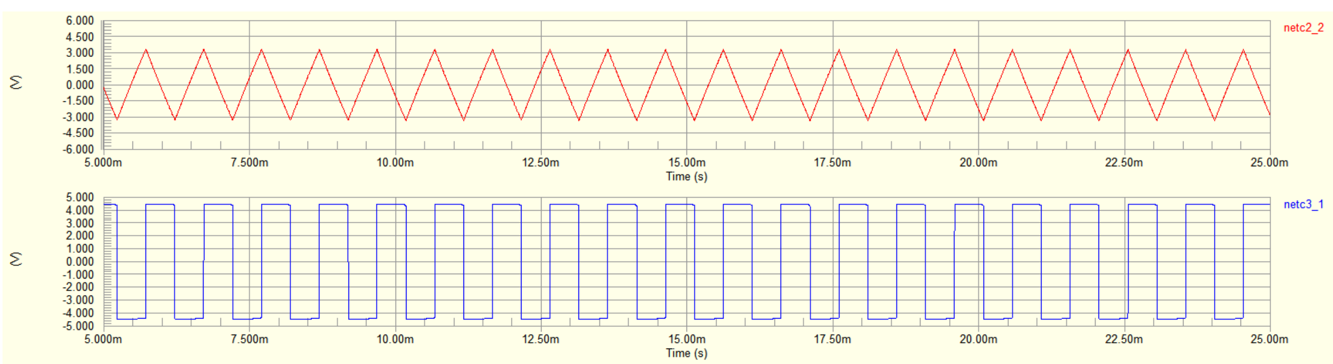


Figure 3 : Altium Simulation of Triangle Wave and the Square wave

Breadboard Testing:

The Sensor fusion unit, PWM unit, PID unit, Triangle Wave generators were assembled on the breadboards and tested via an oscilloscope.

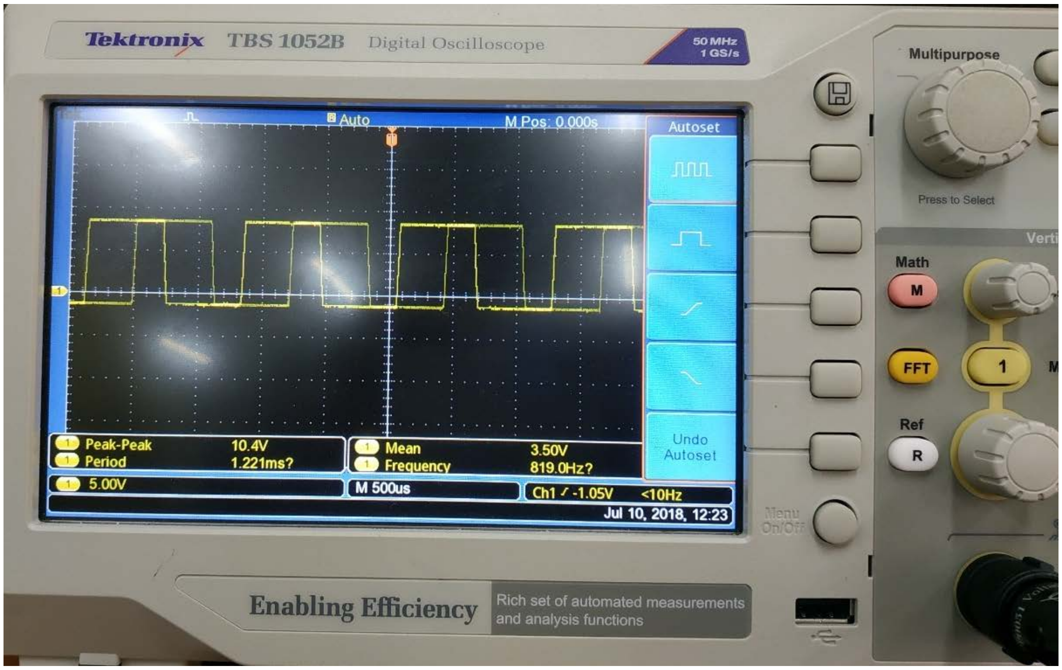


Figure 4: PWM waves generated by the prototypes

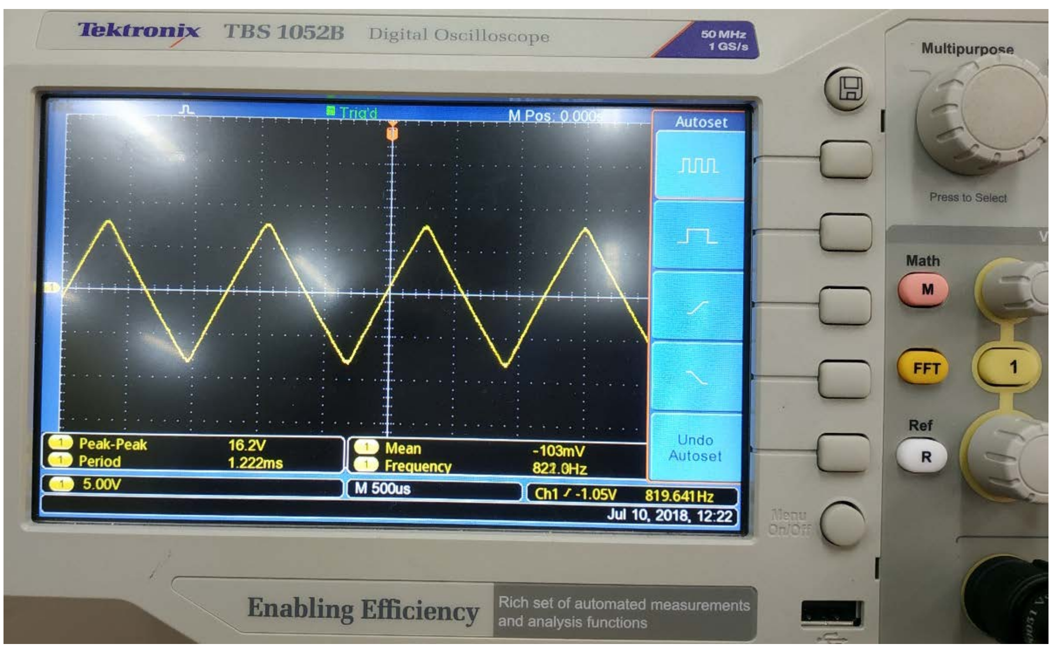


Figure 5: Triangle wave Generated by the prototypes

Testing and results

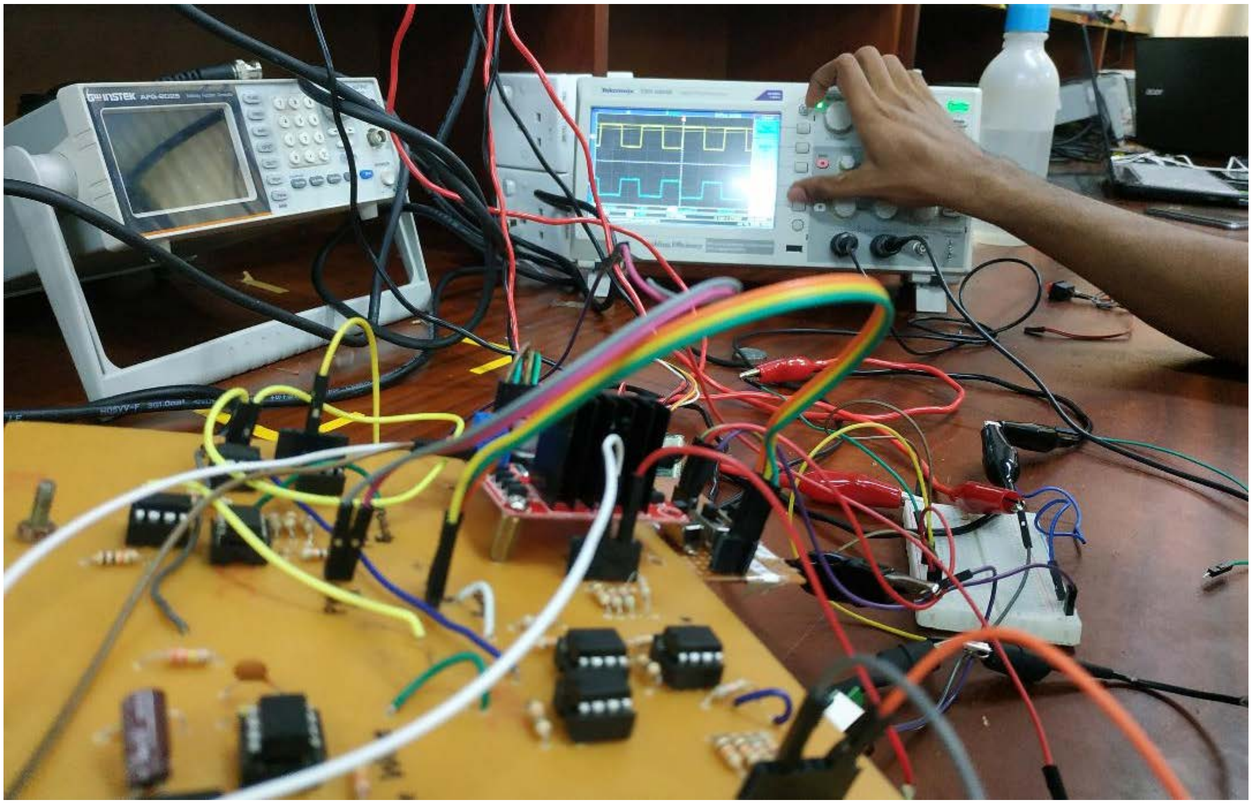


Figure 6: Testing the PCB. PWM output in the background

The robot was able to successfully navigate 80% of the experimental course set at minimum dimensions. It requires one instance of user intervention to reorient itself.

The motors were found to rotate at different speeds for the same PWM duty cycle due to different revolutions per minute.

In corridor larger than 400mm the sensors can detect only one wall at a time due to constrains in sensor range.

Assessment

Strengths

Sensors

The usage of the four analog sensors decreases the potential for collision and allows maneuverability in corridors with concave 270-degree bends.

PID

PID smoothens the robot movement so it can decrease the time taken to complete the task and reduce the probability of collision.

Modularity

Modularity of the robot means that you can easily modify or debug the circuit. This allows more future applications.

Cost-effective

Given the device was made using analog circuit components, the total cost incurred in making the robot is low.

Weaknesses

Weight

The weight of the robot is high due to its heavy power supply units. This means the robot will be slow and the motors will draw more current as more work needs to be done, thus reducing the power efficiency of the robot. The heavy weight also means the robot will not be able to handle inclined and declined surfaces.

Convex Ninety-degree bends.

The robot doesn't have the required functionality to navigate ninety-degree bends

Next steps

As for the next steps it is needed to reduce the weight of the robot, integrate all the units into a solitary PCB, and use a single power supply.

Mapping and Localization: Given that this robot can follow a wall, the robot must be redesigned so that in the indoor applications it doesn't retrace its steps and has the ability for mapping and localization. This is useful for applications such as maze following.

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Appendix A

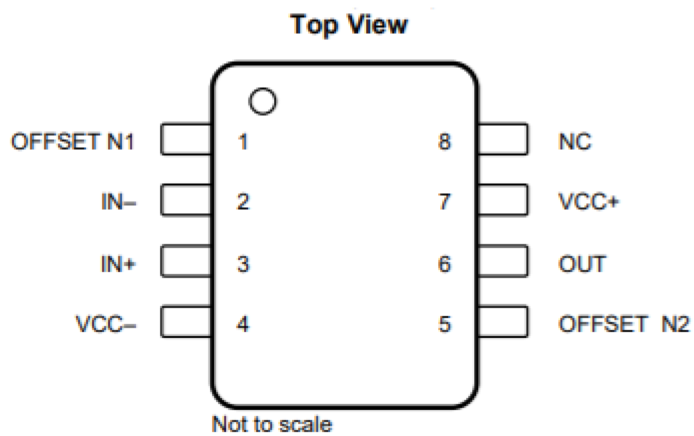
Op-amp

The µA741 device is a general-purpose operational amplifier.

Features

- Short-Circuit Protection
- Offset-Voltage Null Capability
- Large Common-Mode and Differential Voltage Ranges
- No Frequency Compensation Required
- No Latch-Up

Op-amp pin layout



Pin Functions

Pin Functions

PIN NAME	PIN NO.	I/O	DESCRIPTION
IN+	3	I	Noninverting input
IN–	2	I	Inverting input
NC	8	—	No internal connection
OFFSET N1	1	I	External input offset voltage adjustment
OFFSET N2	5	I	External input offset voltage adjustment
OUT	6	O	Output
VCC+	7	—	Positive supply
VCC–	4	—	Negative supply

Recommended Operating Conditions

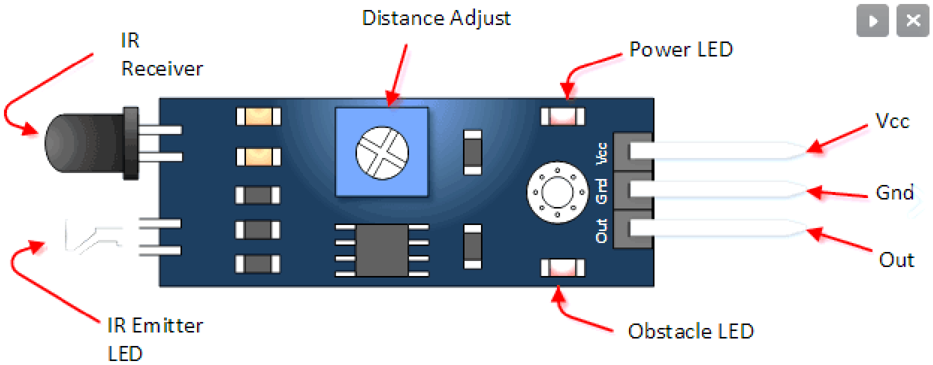
			MIN	MAX	UNIT
V_{CC+}	Supply voltage		5	15	V
V_{CC-}	Supply voltage		–5	–15	V
T_A	Operating free-air temperature	µA741C	0	70	°C

For more information on µA741 refer: http://www.ti.com/lit/ds/symlink/ua741.pdf

Appendix B

Sensor

Sensor Overview



The package has three connection pins:

1. **Vcc** : Power supply of 3-5V DC
2. **Gnd** : Ground reference
3. **Out:** Digital output voltage signal of the sensor

Features

- LM393 Comparator based detection circuit is very stable and accurate
- On board potentiometer sets obstacle detection range
- On board Power LED indicator
- On board Obstacle Detection LED indicator
- 3.0MM mounting hole
- Male headers
- Good Accuracy: Excellent performance in Ambient light conditions

Technical Specifications

Model number	FC-51
Detection angle	35 °
Operating voltage	3.0V - 6.0V
Detection angle	2cm – 30cm (Adjustable using potentiometer)
PCB size	3.1cm(L)*1.4cm(W)
Overall dimensions	4.5cm(L) *1.4cm(W),0.7cm(H)
Current consumption	At 3.3V: ~23 mA At 5.0V: ~43 mA

Further Information:

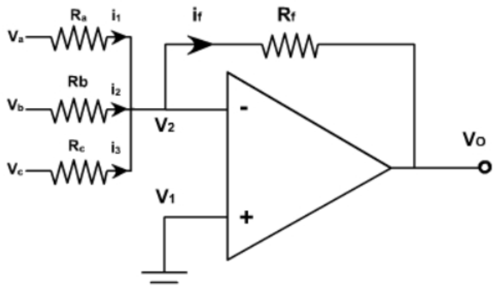
Product review: https://www.youtube.com/watch?v=BVkKX_2JCXk

The Internal Op-amp Datasheet: http://www.ti.com.cn/cn/lit/ds/symlink/lm393-n.pdf

Appendix C

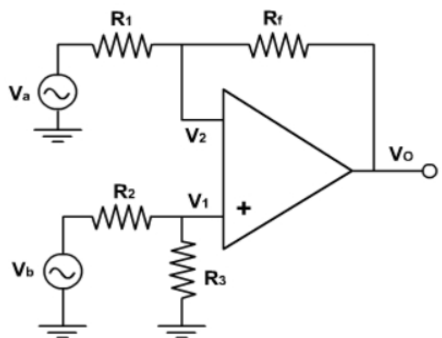
Sensor Fusion Unit

Op-amp Adder.



$$V_o = \left(\frac{R_f}{R_a}V_a + \frac{R_f}{R_b}V_b + \frac{R_f}{R_c}V_c\right)$$

Differential Amplifier

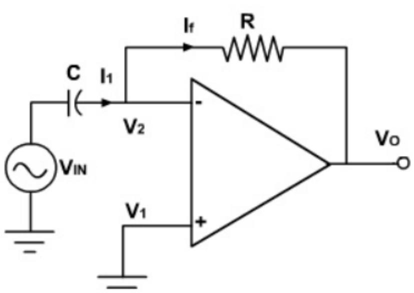


$$\mathrm{V_o} = \frac{\mathrm{R_f}}{\mathrm{R_1}}(-\mathrm{V_a} + \mathrm{V_b})$$

Given that R1= R2 and R3= R4.

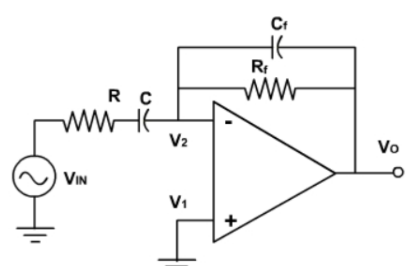
PID Unit

Differentiator

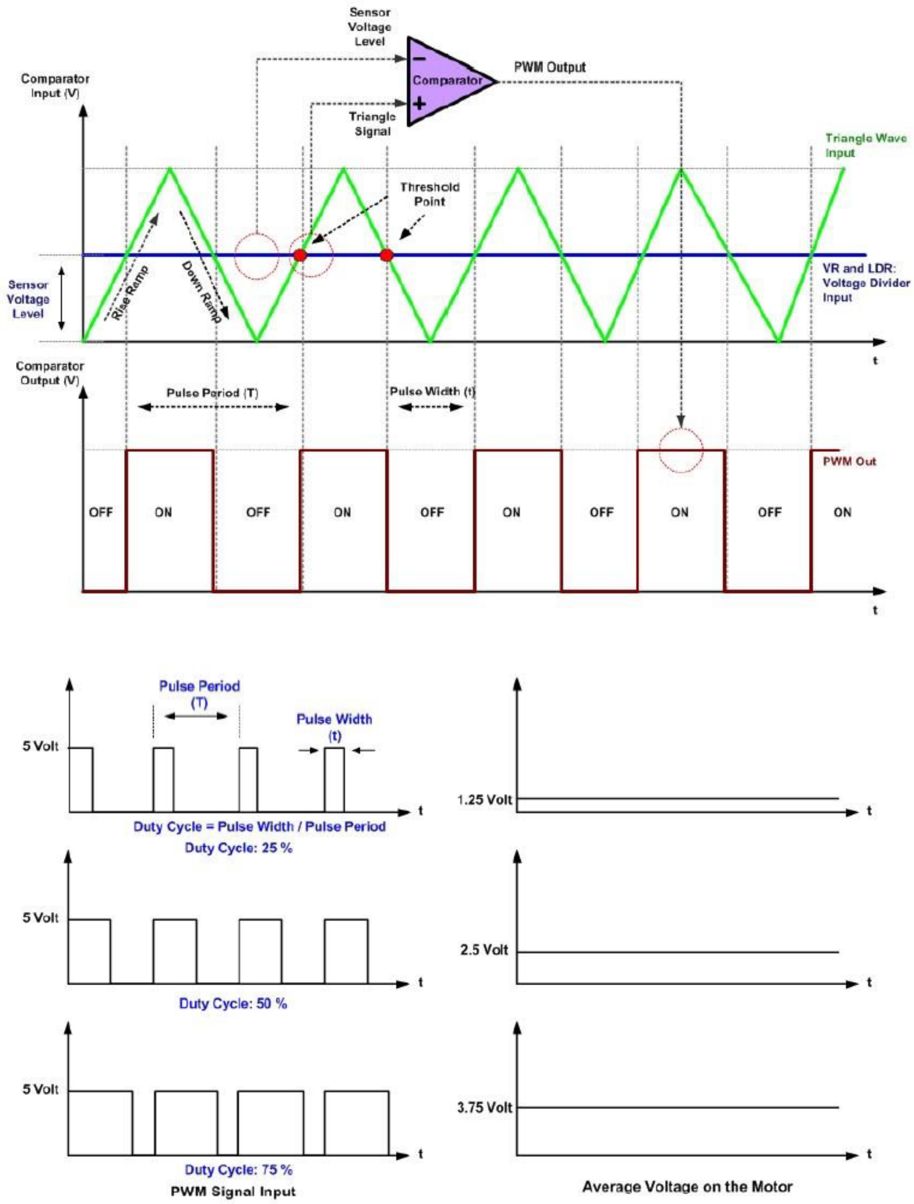


Practical Differentiator

In general, the Op-amp differentiator circuits don't work well as per theory, due to high-frequency gain and noise; Thus, modifications will have to be made. As the frequency changes, the gain changes. At high frequencies, the circuit becomes infected with high frequency noise as the noise gets amplified due to the increase in gain.



PWM Unit



PWM Timing Diagram

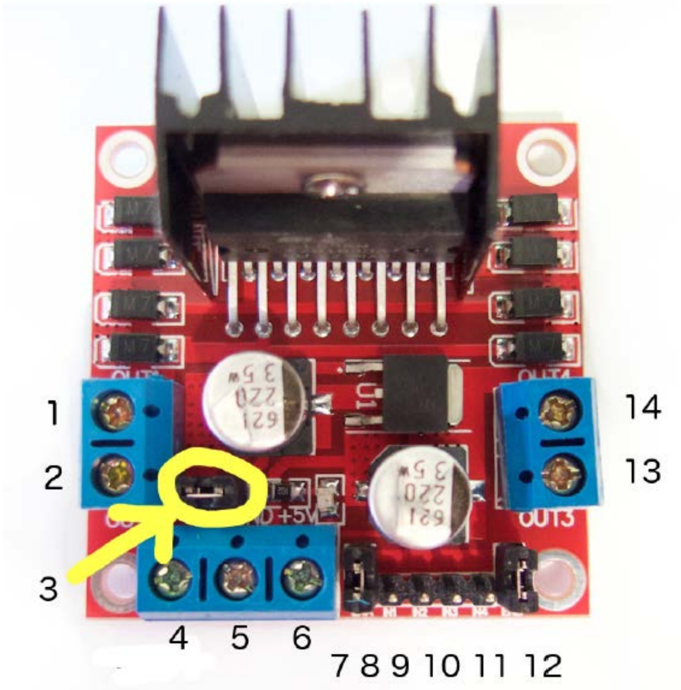
Appendix D

Motor-controller

Introduction

The L298N is a dual H-Bridge motor driver which allows speed and direction control of two DC motors at the same time. The module can drive DC motors that have voltages between 5 and 35V, with a peak current up to 2A.

Pin Layout

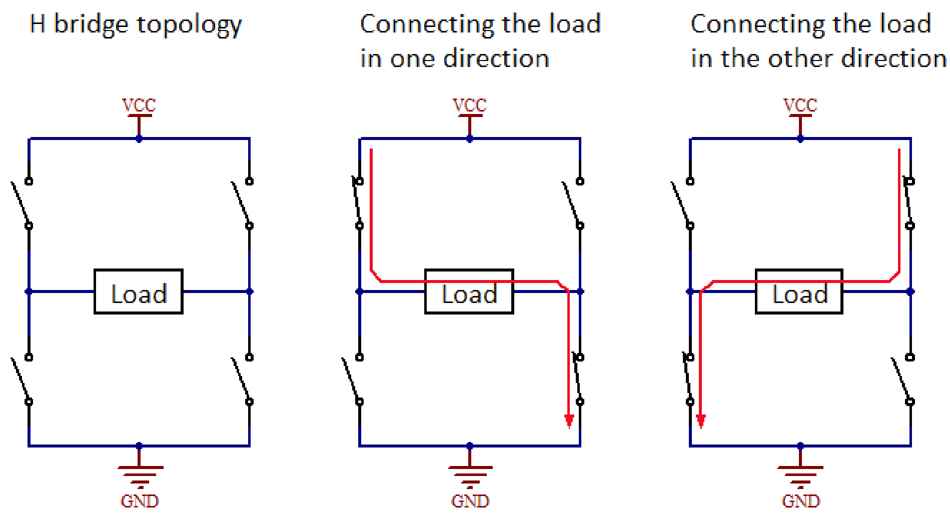


1	**DC motor 1 "+" or stepper motor A+**
2	DC motor 1 "+" or stepper motor A+
3	12V jumper - remove this if using a supply voltage greater than 12V DC. This enables power to the onboard 5V regulator
4	Connect your motor supply voltage here, maximum of 35V DC. Remove 12V jumper if >12V DC
5	GND
6	5V output if 12V jumper in place
7	DC motor 1 enable jumper. Connect to PWM output for DC motor speed control.
8	IN1
9	IN2
10	IN3
11	IN4
12	DC motor 2 enable jumper. Connect to PWM output for DC motor speed control.
13	DC motor 2 "+" or stepper motor B+
14	DC motor 2 "-" or stepper motor B Controlling

H-Bridge DC Motor Control

H-bridge allows you to inverse the direction of current flow through the motor, from which it is possible to change the rotation direction.

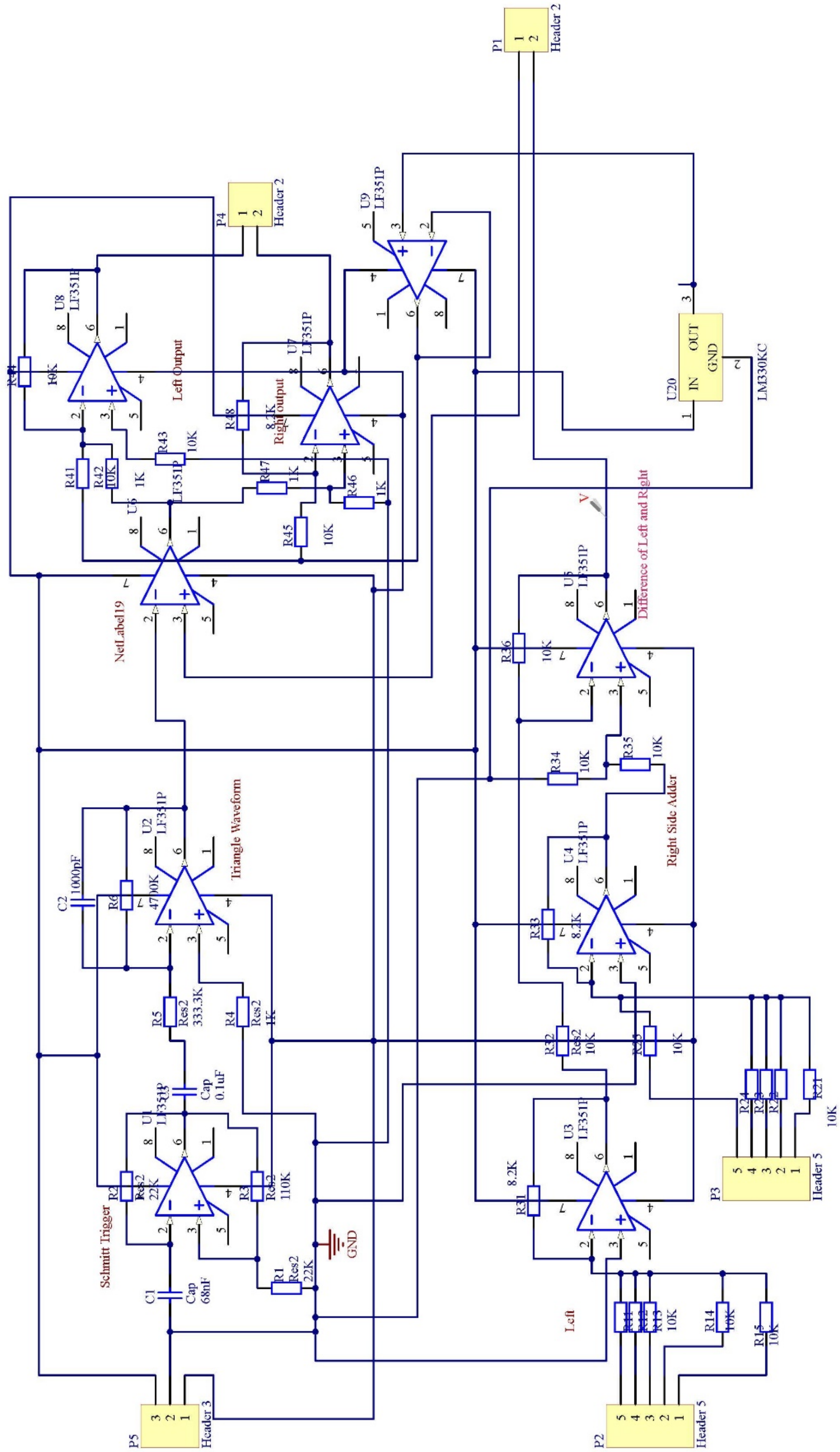
The H-bridge has four switching elements (Transistors and MOSFETs) which allow the control of the current flow.



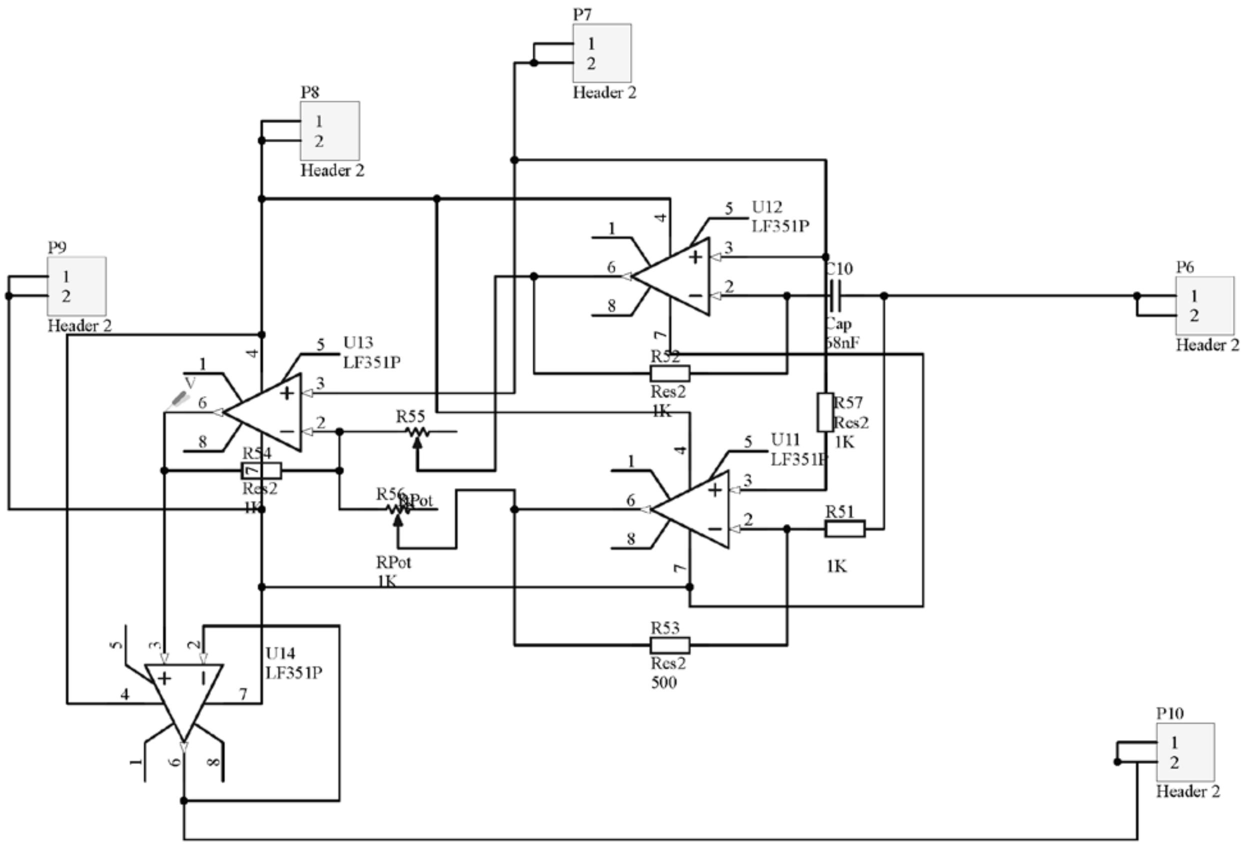
By combining the H-bridge and the PWM, it is possible to have complete control over the DC motor

Appendix E

Schematic of Main PCB



Schematic of PID Circuit



Schematic of Voltage regulator